

AURASeg: Attention Guided Upsampling with Residual Boundary-Assistive Refinement for Robot Drivable-Area Segmentation

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Abstract—Free space ground segmentation is essential to navigate autonomous robots, recognize drivable zones, and traverse efficiently. Fine-grained features remain challenging for existing segmentation models, particularly for robots in indoor and structured environments. These difficulties arise from ineffective multi-scale processing, suboptimal boundary refinement, and limited feature representation. To address this, we propose Attention-Guided Upsampling with Residual Boundary-Assistive Refinement (AURASeg), a ground-plane semantic segmentation framework designed to improve border precision while preserving strong region accuracy. Built on a ResNet-50 backbone, AURASeg introduces (i) a Residual Border Refinement Module (RBRM) that enhances edge delineation through boundary-assistive feature refinement, and (ii) Attention Progressive Upsampling Decoder (APUD) blocks that progressively fuse multi-level features during decoding and additionally, we integrate a (iii) lightweight ASPPLite module to capture multi-scale context with minimal overhead. Extensive experiments on CARL-D, the Ground Mobile Robot Perception (GMRP) dataset, and a custom Gazebo indoor dataset show that AURASeg consistently outperforms strong baselines, with notable gains in boundary metrics. Finally, we demonstrate real-time deployment on a Kobuki TurtleBot, validating practical usability. The code is available at <https://github.com/Narendhiranv04/kobuki-yolop.git>

I. INTRODUCTION

Autonomous robotic navigation relies significantly on semantic segmentation to precisely comprehend the surroundings, enabling safe and efficient motion through both structured and unstructured terrain. Despite remarkable advances in deep learning-based segmentation architectures, challenges remain in feature representation, boundary refinement, and multi-scale learning, which can limit deployment accuracy in real-time robotic applications. Feature extraction and fusion are particularly critical, as they directly affect segmentation accuracy and robustness under viewpoint changes, clutter, and illumination variation.

Traditional segmentation approaches such as DeepLab [1] and DeepLabv3+ [2] introduced Atrous Spatial Pyramid Pooling (ASPP), a methodology for capturing multi-scale contextual information using dilated convolutions with different receptive fields. Subsequent works such as FBRNet [3] improved pyramid-based aggregation by incorporating reinforced spatial pooling to reduce computational overhead while enhancing segmentation quality. In parallel, efficient feature fusion and border/boundary refinement strategies have been explored in architectures such as BiSeNet [4] and

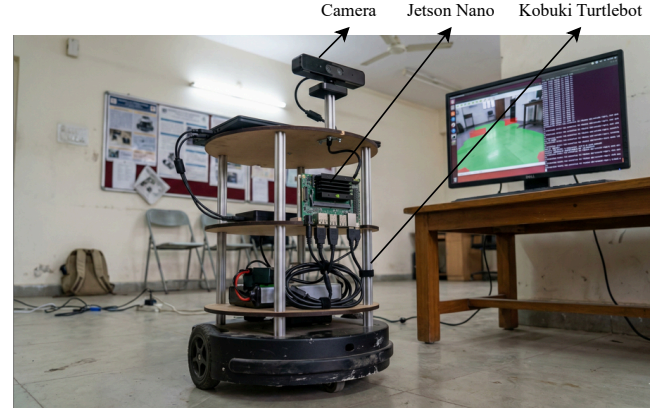


Fig. 1. Real-time drivable-area segmentation on a Kobuki TurtleBot2 using NVIDIA Jetson Nano for onboard inference.

FPANet [5], which leverage residual connections and multi-branch designs to improve edge precision.

However, boundary refinement remains a recurrent issue in segmentation: poor boundary delineation often leads to misclassified pixels near object edges, which reduces the reliability of segmentation-based navigation. For mobile robot perception, such boundary artifacts can propagate to planning as false obstacles or missing free-space, producing overly conservative or unsafe trajectories. This problem is most noticeable in indoor environments and unstructured terrains, where floor segmentation is typically uneven and visually ambiguous. Multi-task learning has also contributed to enhancing feature representation, as YOLOP [6], designed for panoptic prediction, demonstrates how sharing representations across tasks can improve segmentation performance. Nevertheless, since such models are primarily optimized for panoptic outputs, they are not always ideal for pure semantic segmentation where boundary fidelity is critical for reliable robotic navigation. To address these challenges, we propose AURASeg, Attention-Guided Upsampling with Residual Boundary-Assistive Refinement for drivable-area segmentation, integrating:

- 1) *Attention Progressive Upsampling Decoder (APUD)*, an attention-guided decoder that progressively upsamples and fuses multi-scale features to recover fine-grained spatial structure.
- 2) *ASPPLite*, a lightweight multi-scale context module that enriches bottleneck features with minimal computational overhead.
- 3) *Residual Boundary Refinement Module (RBRM)*, a

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boundary-assistive refinement head that leverages a Sobel edge prior and gated residual fusion to sharpen contours and improve boundary-centric metrics.

II. RELATED WORK

Semantic segmentation is a critical component of autonomous navigation, allowing robots to detect and separate drivable areas from obstructions. A backbone’s ability to retain spatial structure while extracting high-level context strongly influences segmentation quality. Recent designs explore improved encoder-decoder alignment and feature selection. Ghost-UNet [7] adopts an asymmetrical encoder-decoder structure to enhance feature alignment, a Dual Stream Encoder Structure is explored in [8], and LCDNet [9] uses gating mechanisms to dynamically select informative features. Across many segmentation families, residual connections [10] remain a common choice to preserve low-level spatial cues and stabilize optimization in deeper networks.

Multi-scale context aggregation is equally important, as it enables a model to combine broad receptive fields with fine-grained spatial detail. ASPP captures multi-scale context via parallel dilated convolutions, improving segmentation performance in challenging scenes. Building on this idea, FBRNet [3] enhances multi-scale extraction using reinforced spatial pooling without incurring excessive computational cost. Beyond purely RGB cues, Depth-Guided DPT [11] incorporated depth-aware segmentation, improving feature extraction in robotic perception scenarios where depth information can be informative. Another efficient approach is S2-FPN [12], which introduced scale-aware strip attention connections to refine multi-scale feature selection and strengthen cross-scale fusion.

Attention mechanisms further complement multi-scale aggregation by selectively emphasizing task-relevant features. Self-attention has been shown to improve representation learning in transformer-based and multimodal fusion segmentation models, especially in complex scenes where long-range dependencies matter. TwinLiteNet [13] introduced dual-attention designs with a focus on lane recognition and outdoor drivable-area segmentation, while attention-based refinement blocks [14] improve feature retention during upsampling and reduce spatial artifacts during resolution recovery.

Finally, boundary precision remains a key determinant of navigation reliability, since misclassified pixels along edges can induce planning errors. BASNet [15] proposed an encoder-decoder residual learning formulation that reinforces edge clarity, and Street Floor Segmentation [16] explored adaptive filtering for refining segmentation masks. Temporal consistency and uncertainty handling have also been studied for mobile robotics. D-Flow [17] introduces Memory-Gated Units (MGUs) to maintain segmentation consistency over sequential frames, and AGSL-Free Driving Region Detection [18] employs uncertainty-aware depth learning to better adapt to low-confidence regions. In contrast to addressing these aspects in isolation, our proposed method consolidates these design choices into a unified framework

tailored for boundary-sensitive free-space segmentation, deployed and validated in a kobuki turtlebot as shown in Figure 1.

III. PROPOSED METHOD

A. Overview of the architecture

As depicted in Figure 2, AURASeg follows an encoder-decoder design tailored for free-space (drivable-area) segmentation in robotic navigation. The input image is first encoded by a ResNet-50 backbone to produce hierarchical feature maps at multiple resolutions. At the bottleneck, ASPPLite aggregates lightweight multi-scale context to improve global scene understanding without incurring heavy computation. The decoder then employs APUD blocks to progressively upsample and fuse encoder features, recovering spatial detail and generating a coarse free-space prediction. Finally, RBRM refines this prediction by injecting boundary-sensitive corrections through gated residual fusion, improving contour alignment while preserving interior-region stability

B. Encoder Backbone

ResNet-50 is used as the encoder backbone, pretrained on ImageNet, to extract hierarchical features for ground segmentation. The encoder produces multi-scale feature maps at progressively lower resolutions (early layers capture edges/textures; deeper layers capture semantics), which are forwarded to the decoder through skip connections for detail recovery during decoder upsampling.

C. ASPPLite Module

ASPPLite is designed to provide multi-scale contextual cues at the bottleneck while keeping computation low for real-time robotic segmentation. It uses four parallel branches: a 1×1 projection branch and three 3×3 atrous convolution branches with dilation rates 1, 6, and 12, each followed by Batch Normalization and ReLU. The $d=1$ branch preserves fine local structure, while $d=6$ and $d=12$ expand the receptive field to incorporate mid- and long-range context, improving robustness to texture changes, illumination variation, and indoor clutter. Unlike standard ASPP variants that add more dilation branches and a global average pooling (GAP) path, ASPPLite omits GAP to avoid spatial collapse and retain boundary-sensitive information, which is critical for ground-plane edges and thin obstacle contours. The outputs of all branches are concatenated to form a richer multi-scale feature tensor, which is then forwarded to the decoder for progressive upsampling and refinement.

D. Attention Progressive Upsampling Decoder (APUD)

Each APUD block fuses a deep, low-resolution semantic feature map $x_{\text{low}} \in R^{B \times C_\ell \times H_\ell \times W_\ell}$ with a shallow, high-resolution detail feature map $x_{\text{high}} \in R^{B \times C_h \times H_h \times W_h}$ to progressively reconstruct the segmentation map. Both inputs are first aligned to a shared embedding dimension C using lightweight 1×1 projections to ensure dimensional consistency before fusion. The semantic branch is then enhanced using Squeeze-and-Excitation (SE) channel attention [19],

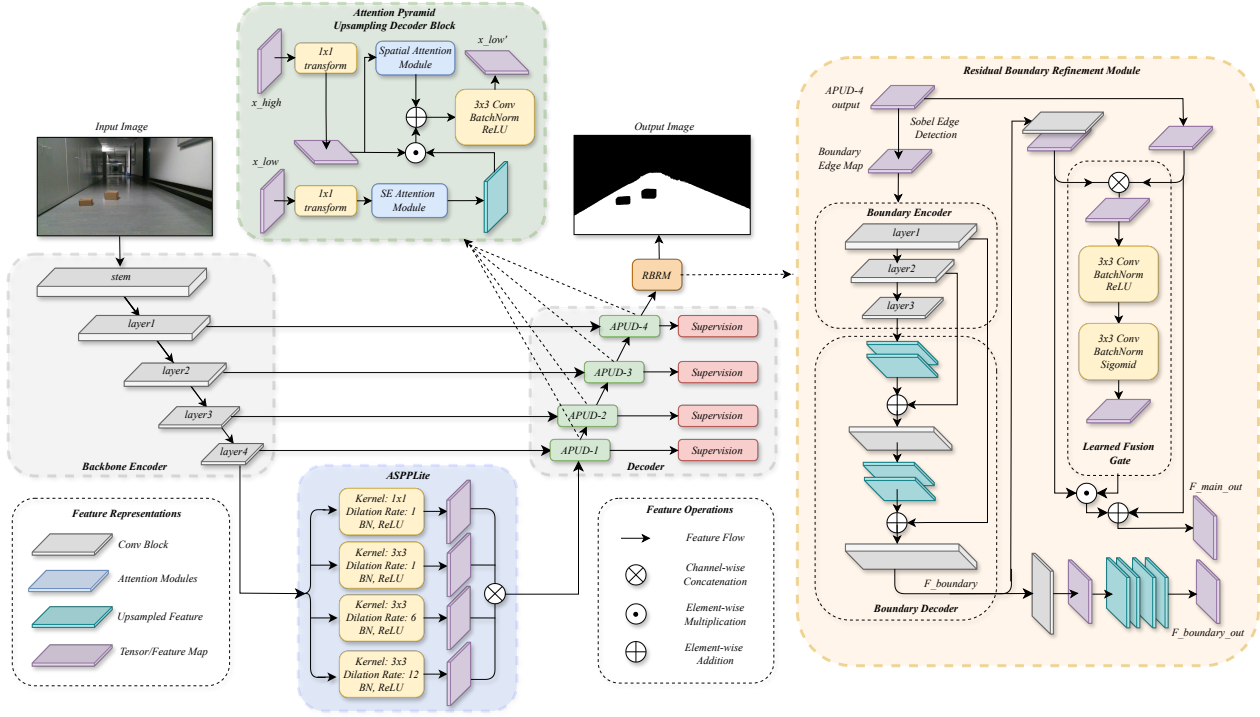


Fig. 2. Overview of the proposed free-space drivable area segmentation encoder-decoder network architecture.

and upsampled via bilinear interpolation to match the spatial resolution of the skip feature. To perform selective fusion, APUD uses an element-wise multiplicative interaction between the upsampled semantic signal and the high-resolution skip feature, acting as a content-dependent gate that suppresses irrelevant textures while retaining boundary-relevant responses. In parallel, the skip branch is refined using a spatial attention [20] mask to emphasize salient regions. The gated fusion and spatially-attended skip are combined through addition and refined using a 3×3 convolution with BN and activation, producing an output feature map at the resolution of x_{high} , i.e., $y \in R^{B \times C \times H_h \times W_h}$.

$$L = T_\ell(x_{\text{low}}), \quad H = T_h(x_{\text{high}}) \quad (1)$$

$$L' = \mathcal{U}(\text{SE}(L)), \quad F = L' \odot H, \quad S = \text{SA}(H) \quad (2)$$

$$y = R(F + S), \quad (3)$$

where $T_\ell(\cdot)$ and $T_h(\cdot)$ are 1×1 Conv-BN-Act projections, $\mathcal{U}(\cdot)$ denotes bilinear upsampling to (H_h, W_h) , $\text{SE}(\cdot)$ is channel-wise attention, $\text{SA}(\cdot)$ is spatial attention, $\odot$ denotes element-wise multiplication, and $R(\cdot)$ is a 3×3 Conv-BN-Act refinement.

E. Residual Boundary Refinement Module (RBRM)

The Residual Boundary Refinement Module (RBRM) is placed after the final APUD stage to improve boundary precision in the predicted free-space mask. While the APUD decoder reconstructs features progressively, small boundary errors can still occur due to upsampling and ambiguity near thin structures or strong texture changes (e.g., floor-wall

transitions and obstacle contours). RBRM adds an explicit boundary-focused refinement branch and merges its cues back into the main decoder features through a gated residual connection. Let the APUD output be $\mathbf{F} \in R^{C \times H \times W}$ (in our setup $C=256$ at $H/4$ resolution).

RBRM first extracts edge-aware features using a lightweight boundary head. The main feature map is projected ($256 \rightarrow 64$) and passed through *fixed Sobel filters* applied per channel to obtain horizontal and vertical responses, which are concatenated and fused with a small learnable convolution to form an edge-sensitive tensor $\mathbf{E} \in R^{64 \times H \times W}$. These edge features are then processed by a compact encoder-decoder pathway: the boundary encoder applies strided convolutions to obtain multi-level boundary representations ($\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3$), and the decoder reconstructs them using bilinear upsampling. At each decoding stage, the upsampled features are concatenated channel-wise with the corresponding encoder features and refined by convolution blocks, producing boundary features $\mathbf{B} \in R^{64 \times H \times W}$. The boundary features are projected to match the main feature dimensionality, $\mathbf{P} = \phi(\mathbf{B}) \in R^{C \times H \times W}$, and fused into the main stream through a learned gate:

$$\mathbf{G} = \sigma(\text{Conv}_{1 \times 1}([\mathbf{F}, \mathbf{P}])) , \quad \mathbf{F}_{\text{ref}} = \mathbf{F} + \mathbf{G} \odot \mathbf{P}, \quad (4)$$

where $[\cdot, \cdot]$ denotes channel concatenation, $\sigma(\cdot)$ is the sigmoid function, and $\odot$ is element-wise multiplication. This gated formulation injects boundary cues only where needed while preserving stable interior regions. In addition to refined features $\mathbf{F}_{\text{ref}}$ for the final segmentation head, RBRM also produces an auxiliary boundary map via a 1×1 prediction

TABLE I
OVERVIEW OF DATASETS USED FOR DRIVABLE-AREA SEGMENTATION.

Dataset	Environment	Train	Val	Test	Total
Gazebo (ours)	Indoor lab corridors (simulation)	2483	294	420	3197
GMRPD [21]	Outdoor sidewalks, plazas, squares	616	74	110	800
CARL-D [22]	On-road driver-view scenes for autonomous driving	9000	2400	2300	14700

TABLE II
TRAINING CONFIGURATION FOR AURASEG V4-R50.

Hyperparameter	Value
<i>Optimization & schedule</i>	
Input resolution	384×640
Encoder backbone	ResNet-50 (ImageNet-pretrained)
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)
Weight decay	0.01
Learning rate (enc/dec)	1×10^{-4} / 1×10^{-3}
Scheduler	Cosine annealing
Epochs	50 (early stop patience: 10)
Batch size	8
Mixed precision	FP16 (AMP)
<i>Loss functions</i>	
Main loss (region)	$0.5 \mathcal{L}_{\text{Focal}} + 0.5 \mathcal{L}_{\text{Dice}}$
Boundary loss	$0.2 \mathcal{L}_{\text{BCE}}$
Auxiliary loss	$0.1 (\mathcal{L}_{\text{Focal}} + \mathcal{L}_{\text{Dice}}) \times 4$
Augmentation	Prob. Parameters
Horizontal flip	0.5 –
Geometric (shift/scale/rotate)	0.5 shift=0.1, scale=0.1, rot= $\pm 15^\circ$
Brightness + contrast	0.3 $\Delta b = \pm 0.2$, $\Delta c = \pm 0.2$
Gaussian noise	0.2 var_limit=(10, 50)
Normalize	1.0 ImageNet mean/std

layer on **B**, which can be used during training to encourage boundary-aware refinement.

F. Multi-Loss Supervision and Training

As summarized in Table II, we train AURASeg at an input resolution of 384×640 using an ImageNet-pretrained ResNet-50 encoder. We optimize the network with AdamW to stabilize fine-tuning of the pretrained backbone while enabling faster adaptation of the newly introduced modules (ASPP Lite, APUD, RBRM, and the segmentation head) via differential learning rates: 1×10^{-4} for the encoder and 1×10^{-3} for the decoder-side modules. A cosine-annealing schedule is applied for up to 50 epochs with early stopping (patience = 10) based on validation performance. Training is conducted with batch size 8 and mixed precision (FP16/AMP) on a single NVIDIA GeForce RTX 5060 GPU. All baseline models were trained from scratch under identical conditions: same input resolution (384×640), augmentation pipeline, batch size, optimizer (AdamW), and training budget (100 epochs) to ensure fair comparison.

AURASeg is trained with three complementary supervision signals: (i) a region loss on the main prediction, (ii) deep supervision on intermediate APUD outputs, and (iii) a boundary-specific loss for the RBRM output. Let Y denote

TABLE III
ABLATION STUDY OF DIFFERENT CONFIGURATIONS.

Variant	Params (M)	GFLOPs	Acc.	B. Acc.
V1: Base model	27.90	31.2	0.9928	0.7804
V2: V1 + ASPP-Lite	34.32	33.9	0.9928	0.7931
V3: V2 + APUD	34.77	34.6	0.9943	0.8224
V4: V3 + RBRM (proposed model)	44.48	36.8	0.9946	0.8504

the ground-truth segmentation mask and $\hat{Y}$ the main logits (bilinearly resized to the ground-truth resolution before loss computation). Let $\hat{Y}^{(k)}$ denote the k -th auxiliary logits from the APUD decoder ($k \in \{1, \dots, 4\}$), also resized to the ground-truth resolution. The overall objective is:

$$\mathcal{L} = \mathcal{L}_{\text{main}} + \lambda_{\text{bnd}} \mathcal{L}_{\text{BCE}} + \sum_{k=1}^4 \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}^{(k)}, \quad (5)$$

where $\lambda_{\text{bnd}} = 0.2$ and $\lambda_{\text{aux}} = 0.1$ (Table II).

Main region loss. We combine Focal loss and Dice loss to balance class-imbalance robustness with overlap-driven optimization:

$$\mathcal{L}_{\text{main}} = 0.5 \mathcal{L}_{\text{Focal}}(\hat{Y}, Y) + 0.5 \mathcal{L}_{\text{Dice}}(\hat{Y}, Y). \quad (6)$$

Auxiliary deep supervision. Each intermediate APUD output is supervised with the same region criterion for stages $k \in \{1, \dots, 4\}$:

$$\mathcal{L}_{\text{aux}}^{(k)} = \mathcal{L}_{\text{Focal}}(\hat{Y}^{(k)}, Y) + \mathcal{L}_{\text{Dice}}(\hat{Y}^{(k)}, Y). \quad (7)$$

Boundary loss for RBRM. The RBRM predicts a boundary-logit map $\hat{B}$ (upsampled to ground-truth resolution). The target B is derived from Y via a thin edge band (e.g., morphological gradient). Let $p_i = \sigma(\hat{B}_i)$. The boundary loss is:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N \left(B_i \log p_i + (1 - B_i) \log(1 - p_i) \right). \quad (8)$$

Here, $\sigma(\cdot)$ is the sigmoid and N is the number of pixels, encouraging region consistency and sharp boundary alignment for drivable-area segmentation in robotic scenes.

IV. RESULTS

A. Datasets

As summarized in Table I, we use a synthetic indoor Gazebo dataset collected for robot navigation, the RGB-D GMRPD [21] dataset representing real outdoor traversal

scenes for ground robots (sidewalks/plazas), and CARL-D [22] representing road-scene segmentation in autonomous driving.

B. Performance Analysis

1) *Ablation Study*: Table III presents a step-wise ablation of our proposed model. Accuracy (Acc.) denotes pixel-wise classification accuracy and Boundary Accuracy (B. Acc.) is the Boundary F1 score, calculated on boundary pixels extracted via morphological dilation-erosion. Adding ASPPLite (V2) modestly increases parameters/GFLOPs while improving boundary accuracy ($0.7804 \rightarrow 0.7931$), indicating better multi-scale context without degrading region accuracy. Introducing the proposed APUD decoder (V3) yields the largest jump in boundary accuracy ($0.7931 \rightarrow 0.8224$) and improves overall accuracy ($0.9928 \rightarrow 0.9943$), highlighting the benefit of attention-guided skip fusion during reconstruction. Finally, incorporating the boundary refinement stage (V4, proposed) further strengthens boundary accuracy to 0.8504 with a moderate compute increase, confirming that explicit boundary-focused refinement complements ASPPLite and APUD modules.

2) *Metric Evaluation*: Table IV summarizes the evaluation of all benchmark methods on the MIX (Gazebo+GMRPD) validation set and CARL-D test set. We export AURASeg to ONNX and deploy using TensorRT on a Jetson Nano 4GB. Performance is reported with batch size 1 at 384×640 input resolution. Latency/FPS is measured end-to-end (resize + normalization, TensorRT inference, and mask upsampling/argmax postprocessing), averaged over 500 runs after 50 warmup iterations.

Boundary Evaluation Protocol. We compute Boundary IoU (BIOU) and Boundary F1 (BF1) by extracting a boundary band using morphological gradient operations (3×3 kernel), then applying $k=2$ dilation iterations to account for minor spatial misalignments while penalizing gross boundary errors.

MIX Dataset (Gazebo+GMRPD). AURASeg achieves the strongest boundary localization, reaching BIOU=0.8124 and BF1=0.8905, improving over the best baseline (UPerNet-R50: BIOU=0.7863, BF1=0.8738) by 3.3% and 1.9% respectively. These gains indicate sharper border alignment, critical for drivable-area segmentation in cluttered

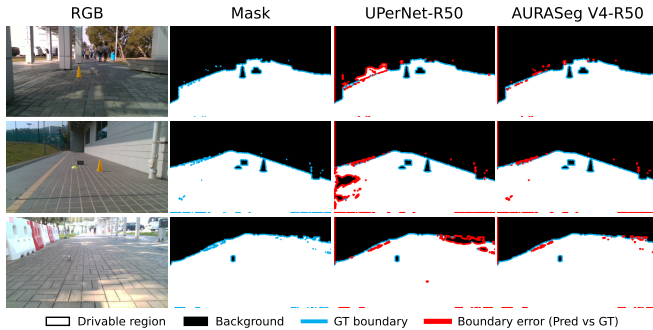


Fig. 3. Qualitative comparison with boundary-error overlays (red) and ground truth boundary (blue) for UPerNet and AURASeg models

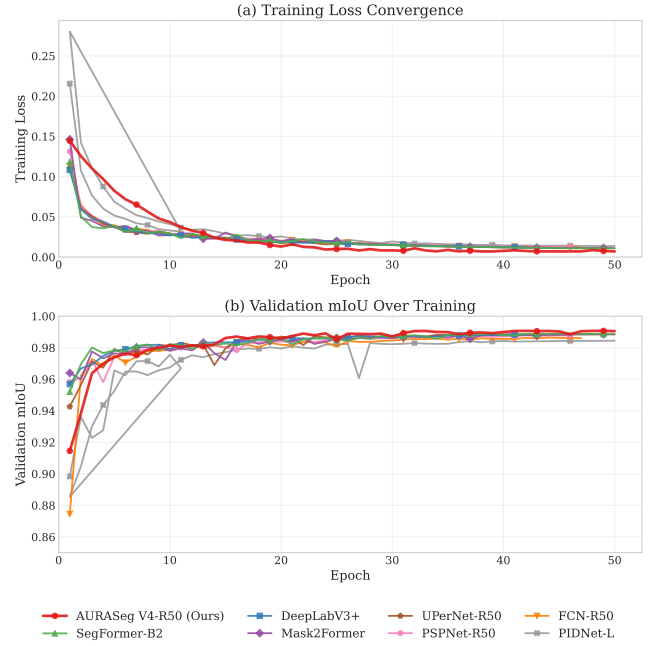


Fig. 4. Training plots of AURASeg and baselines, showing stable convergence and improved validation performance for the proposed method.

indoor environments. Region-level metrics are equally strong ($\text{IoU}_{\text{drv}}=0.9897$, $\text{F1}=0.9948$), confirming that boundary refinement complements interior-region stability.

CARL-D Dataset. On the CARL-D test set, AURASeg again achieves the best performance: $\text{IoU}_{\text{drv}}=0.8041$, $\text{F1}=0.8914$, $\text{BIOU}=0.0484$, and $\text{BF1}=0.0683$. The lower absolute boundary values stem from CARL-D’s smooth polygon annotations, which produce fewer boundary pixels (~ 200 vs ~ 5000 in MIX).

Fig. 3 qualitatively compares RGB inputs, ground truth, and predictions across representative samples, with boundary error maps highlighting that AURASeg produces noticeably cleaner and better-aligned free-space contours than competing methods, particularly around challenging edges. As shown by the training curves in Fig. 4, the proposed model converges rapidly and maintains a stable validation plateau, supporting reliable optimization under the unified training protocol.

V. CONCLUSION

Accurate drivable-area segmentation is essential for reliable navigation of mobile robots in both structured indoor corridors and unstructured outdoor walkways. In this work, we present AURASeg, a lightweight encoder-decoder segmentation framework that couples efficient multi-scale context aggregation (ASPPLite) with an *Attention-guided Progressive Upsampling Decoder* (APUD). To explicitly address the key limitation of blurred or misaligned borders in free-space segmentation, we further introduced a *Residual Boundary Refinement Module* (RBRM) that learns boundary-sensitive features and injects them back into the main stream through a learned gated residual fusion, improving edge

TABLE IV
UNIFIED EVALUATION OF PROPOSED AURASEG MODEL ON MIX (GAZEBO+GMRPD) VALIDATION SET AND CARL-D TEST SET.

Model	MIX (Gazebo+GMRPD) – Validation ($\uparrow$)						CARL-D – Test ($\uparrow$)					
	IoU	F1	BloU	BF1	Prec.	Rec.	IoU	F1	BloU	BF1	Prec.	Rec.
FCN	0.9857	0.9928	0.6502	0.7789	0.9919	0.9936	0.7735	0.8723	0.0298	0.0448	0.8457	0.9006
PSPNet	0.9870	0.9935	0.7639	0.8589	0.9941	0.9929	0.7835	0.8786	0.0422	0.0624	0.8594	0.8987
DeepLabV3+	0.9875	0.9937	0.7799	0.8700	0.9938	0.9936	0.8012	0.8896	0.0416	0.0597	0.8481	0.9354
UPerNet	0.9879	0.9939	0.7863	0.8738	0.9948	0.9931	0.7965	0.8868	0.0240	0.0371	0.8333	0.9475
SegFormer	0.9885	0.9942	0.7763	0.8683	0.9958	0.9927	0.7675	0.8685	0.0184	0.0316	0.8613	0.8758
Mask2Former	0.9881	0.9940	0.7740	0.8661	0.9955	0.9925	0.7721	0.8714	0.0183	0.0298	0.8284	0.9190
PIDNet	0.9835	0.9917	0.6334	0.7656	0.9912	0.9922	0.7979	0.8876	0.0408	0.0616	0.8580	0.9192
AURASeg (Ours)	0.9897	0.9948	0.8124	0.8905	0.9959	0.9937	0.8041	0.8914	0.0484	0.0683	0.8534	0.9330

Note: Best results per metric within each dataset are highlighted in bold. Boundary metrics are computed using boundary dilation = 2.

alignment without degrading interior-region stability. On the robot-centric mix of Gazebo+GMRPD datasets, our method preserves strong region accuracy while improving boundary-centric scores, and its robustness is further demonstrated on the CARL-D road-scene dataset. Future work will extend to strengthen performance in visually ambiguous boundary cases like shadows, reflections and thin structures, where free-space contours are hard to resolve.

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